

Package #154

Implement lead-free solder joint reinforcement techniques that enhance thermo-mechanical fatigue resistance without adding significant mass or compromising heat dissipation

Technology Maturity: ENGINEERING_DEFINITION

Transfer Posture: Sponsored Validation

Problem: Documented aerospace reliability problem: BGA solder joints in satellite electronics accumulate thermo-mechanical fatigue damage from repeated thermal cycling between sunlight and eclipse (~-150C to +120C per orbit, ~5500 cycles/year in LEO), eventually cracking and producing intermittent or permanent open circuits. The failure is field-documented across aerospace electronics heritage programs and is a dominant driver of avionics derating and redundancy mass. This description is qualitative problem-domain context from the engineering literature; no specific incident statistics are asserted here (the engine's own retrieval must find and custody the evidence).

Mechanism: measures to improve the reliability of solder joints

Contents:

- 01 - Executive Technology Brief
- 02 - Engineering Technology-Transfer Dossier
- 03 - Buyer Decision Card
- 04 - Evidence Summary
- 05 - Transfer Manifest

Status: ENGINEERING_DEFINITION. Not physically validated.